

Disaster-resilient traffic management & network planning:

Cases of the 2016 & 2026 Kumamoto earthquakes

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Stories

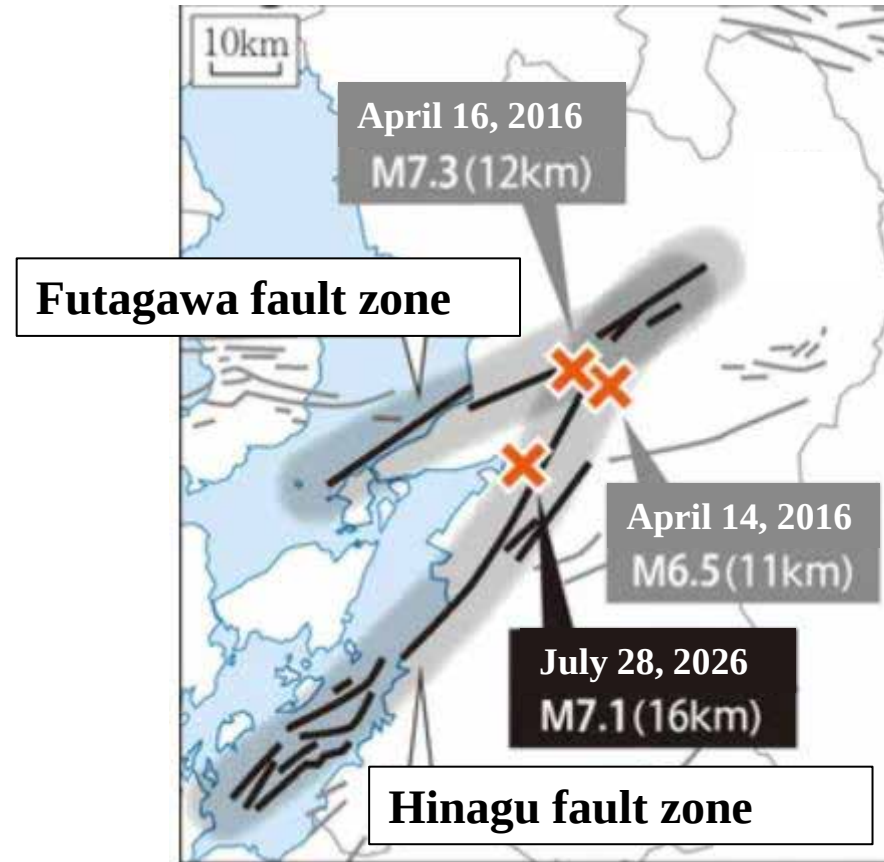
- **Mega-earthquakes** struck Kumamoto on **April 14 & 16, 2016**
- Kumamoto University team supported recovery planning for 10+ years
- Sharing key lessons learned

- **Another mega-earthquake** hit Kumamoto on **July 28, 2026** (1 week ago)
- Severe **congestion** caused by expressway & Shinkansen closures
- Traffic management in collaboration with government
- Expressway damage & recovery status (based on site visit on Aug 2)
- **Ongoing** stories

The 2016 & 2026 Kumamoto earthquakes



Japan Median Tectonic Line



Inland earthquakes within a landplate, causing severe localized shaking directly above the epicenter

Rescuers race to find survivors following massive Kumamoto quake



The 2026 earthquake triggered a large explosion at a popular shopping mall

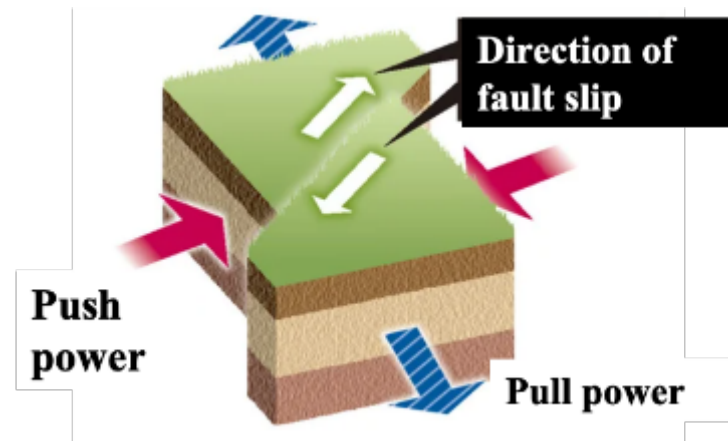
<https://youtube.com/shorts/W4LMVkn0K6w?si=UhWAdNmpmWLiZSzc>

<https://www.japantimes.co.jp/news/2026/07/29/japan/kumamoto-quake-explosion-mall/>

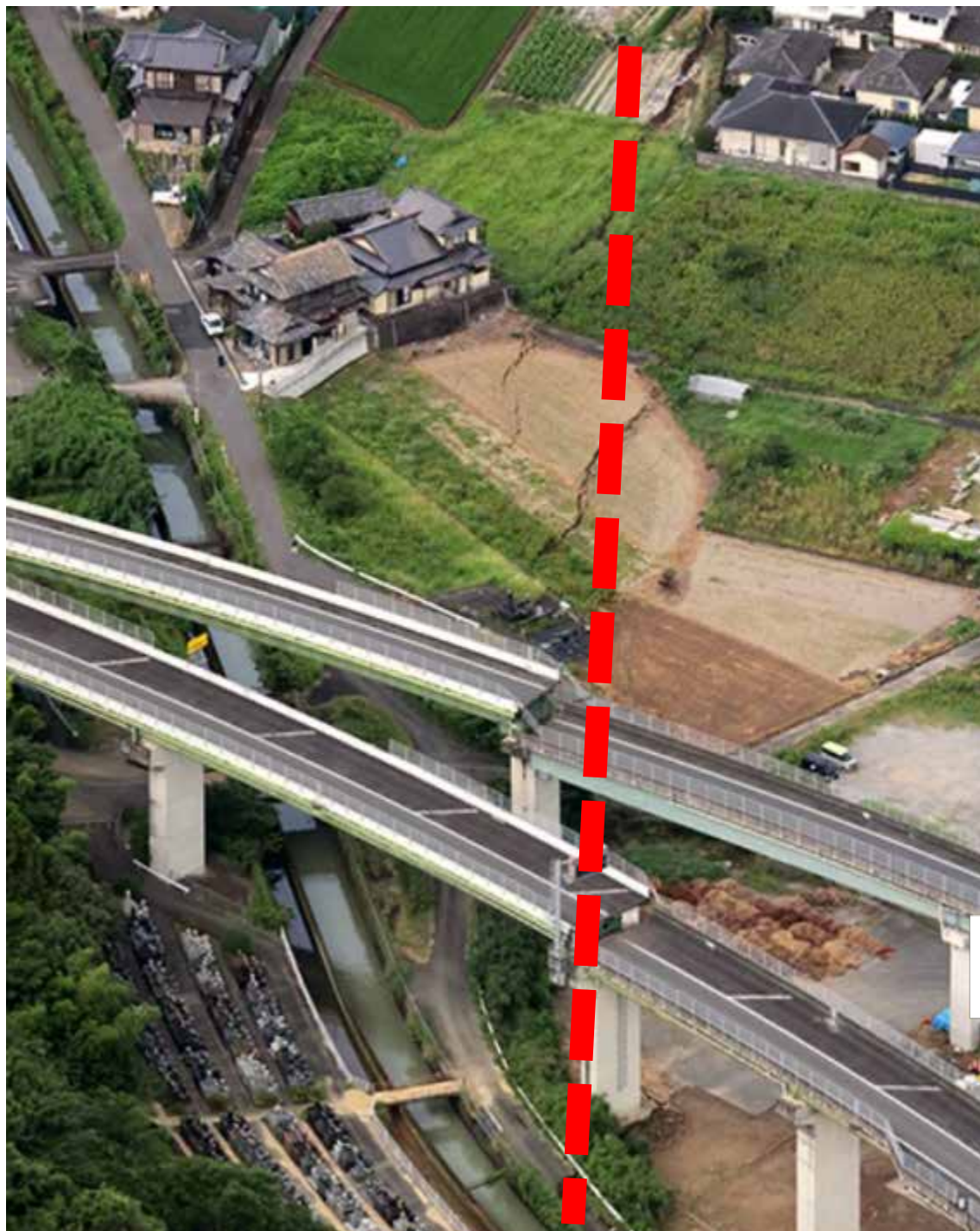


Displaced road

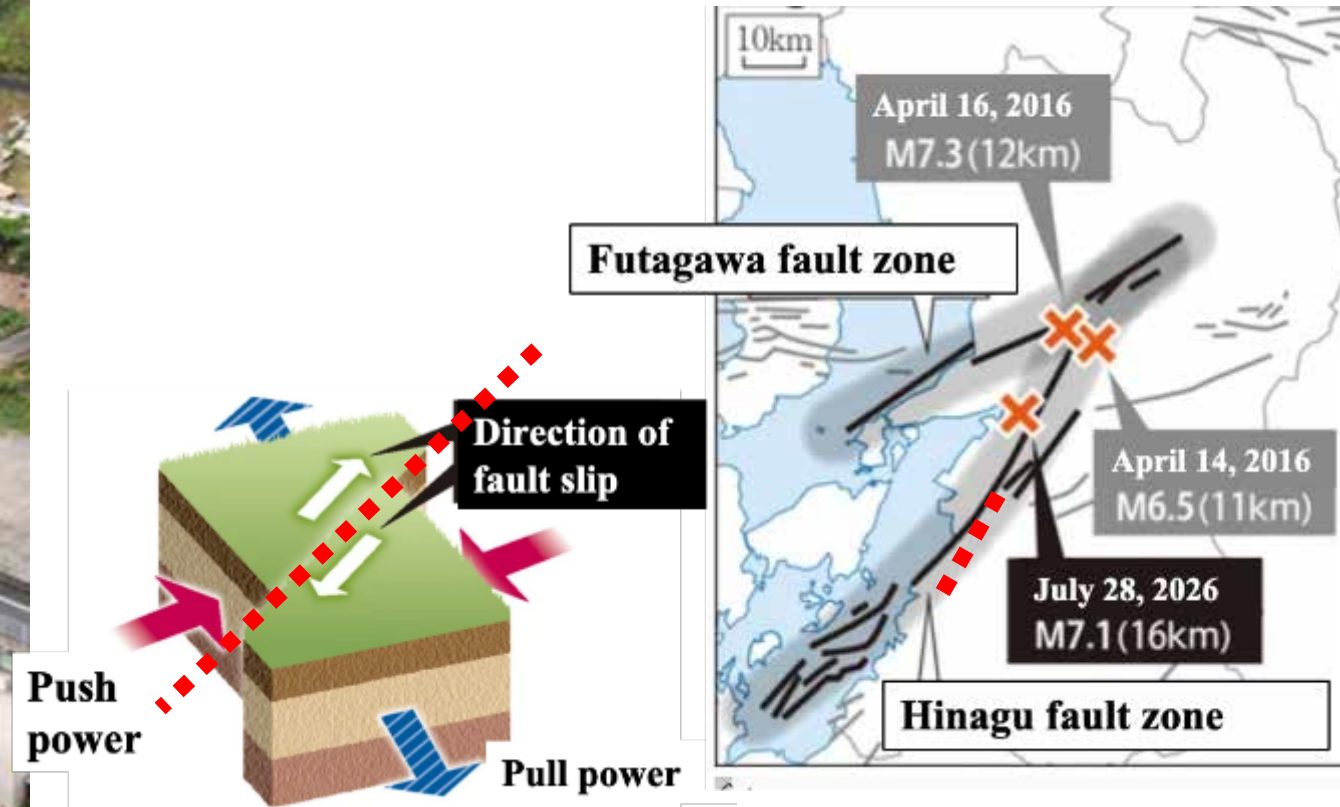
near the Yatsushiro Junction on the
Minami-Kyushu Expressway after the
2026 Kumamoto earthquake



<https://www.asahi.com/ajw/articles/photo/83432749>



A fault line runs along the displaced road



<https://www.asahi.com/ajw/articles/photo/83432749>



Photo Taken by me
August 2, 2026



Photo Taken by me
August 2, 2026



Japanese civil engineers responsibly formulate & execute rapid recovery plans

August 2, 2026

Extreme Heat Hindering Recovery & Evacuation Efforts



- Kumamoto city hits record-high **40.3 ° C** on August 3
- **Compound disaster**
- 2016 Spring vs. 2026 Summer: extreme heat complicates evacuation (including in-car)



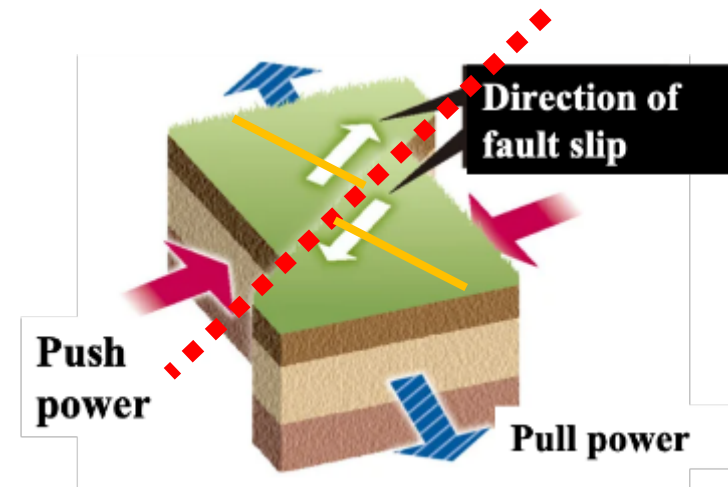
Building damage
in Yatsushiro
Mainly old one

August 2, 2026
Photo taken by me



Fault displacement offset the centerline by ~90 cm

(Toyono, Uki)





Congestion

Grid locks in Kumamoto Center area

18:00 July 28, 2026 on Google Map
(Earthquake occurred at 16:26)

Possibly due to **immediate** post-earthquake **homebound trips**



Heavy traffic congestion in the affected area

Obstacles for evacuation, emergency, & support services



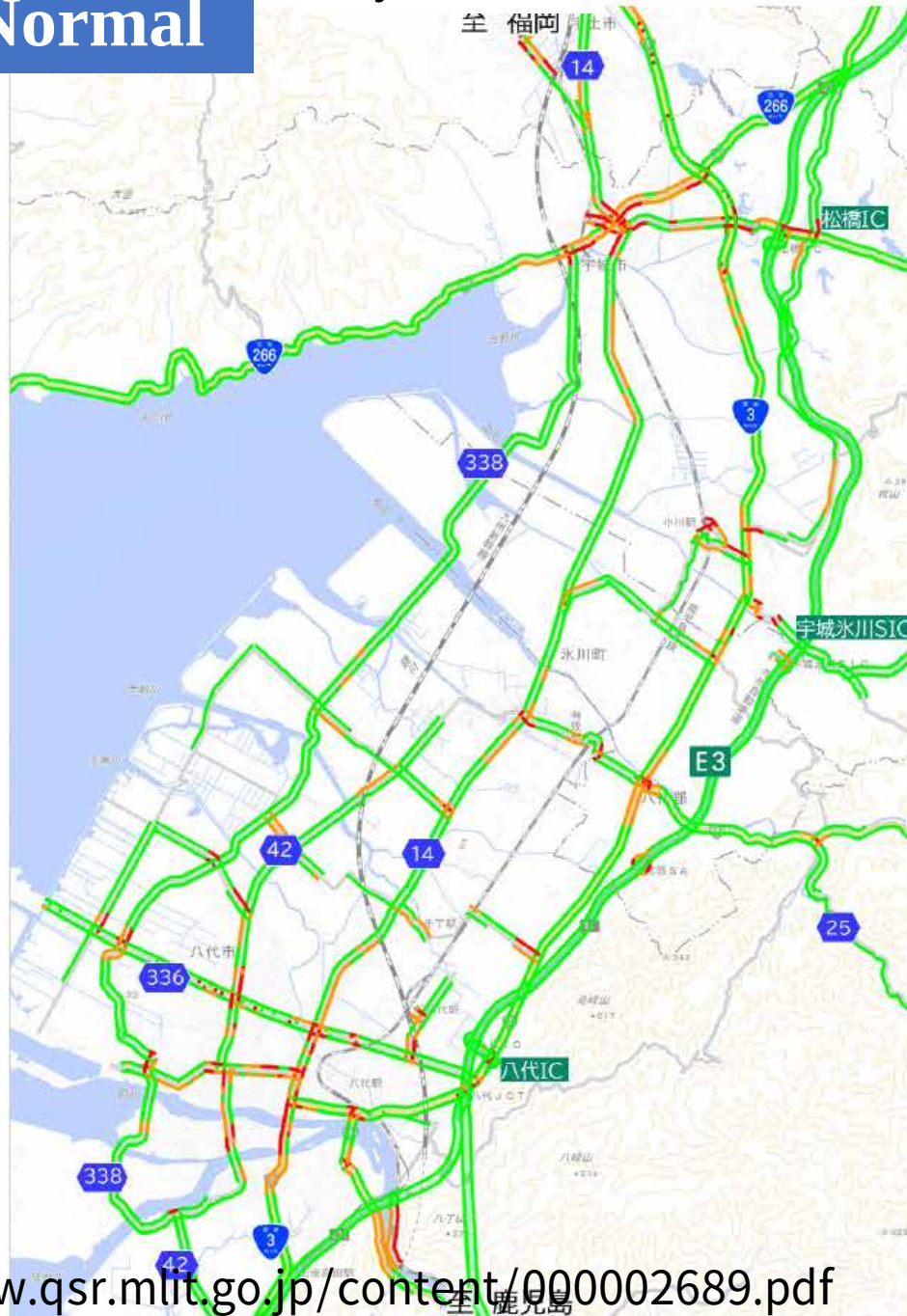
**Top headline in the local newspaper
(August 1, 2026)**

<https://news.yahoo.co.jp/articles/75a6288775c935bf1f68dc60e94cf30d82284584>

Normal

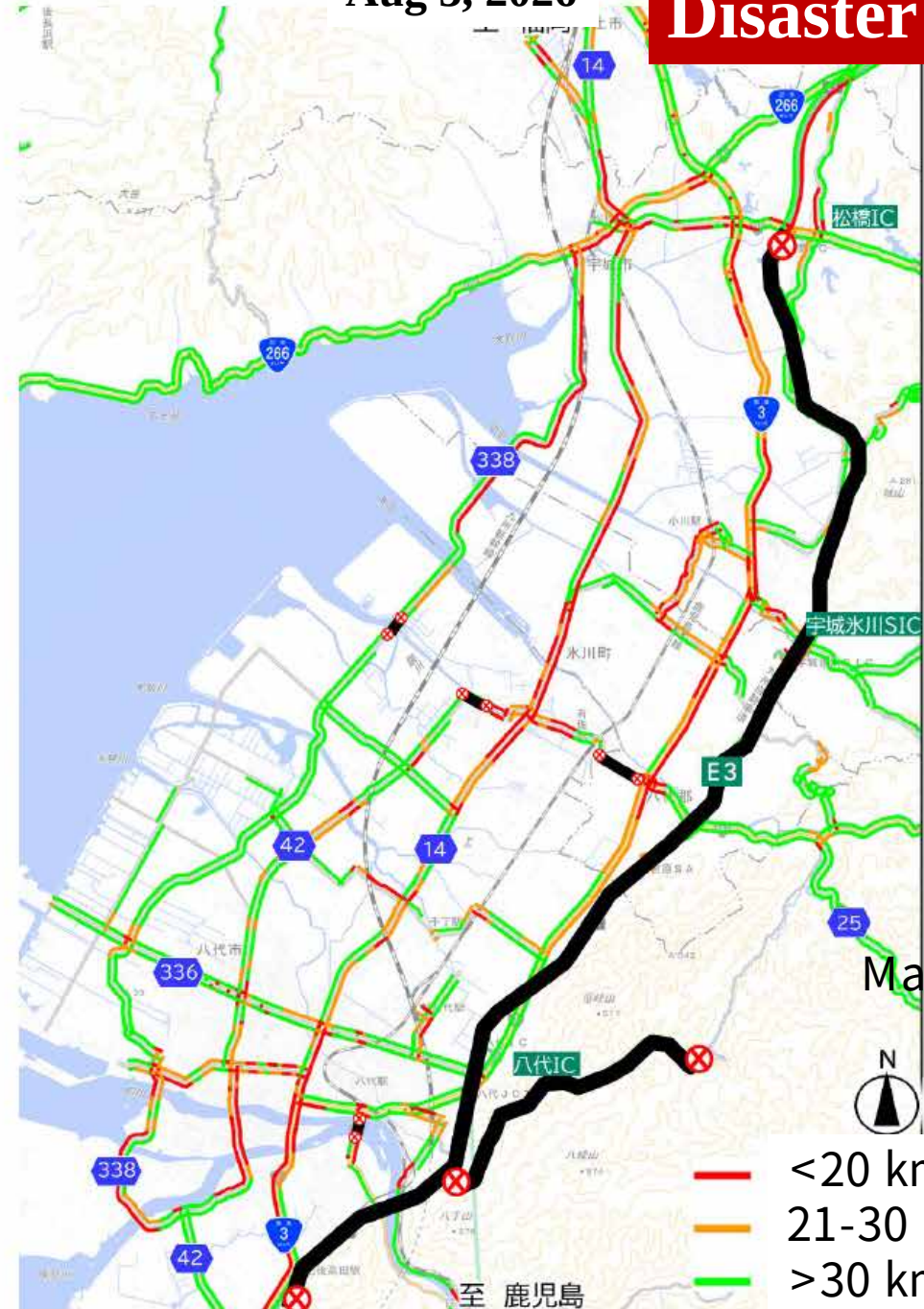
July 27, 2026

7:00~9:00



Aug 3, 2026

Disaster



Map by MLIT

<20 km/h
21-30 km/h
>30 km/h

Expressway Damage & Closure

Shinkansen Rail Damage

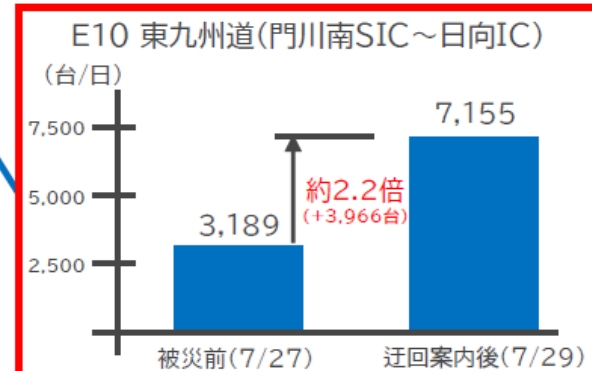
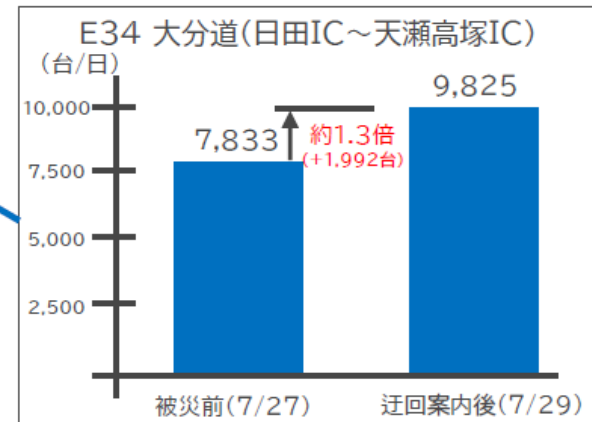
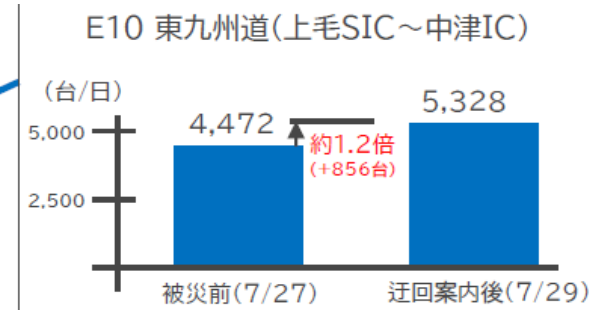
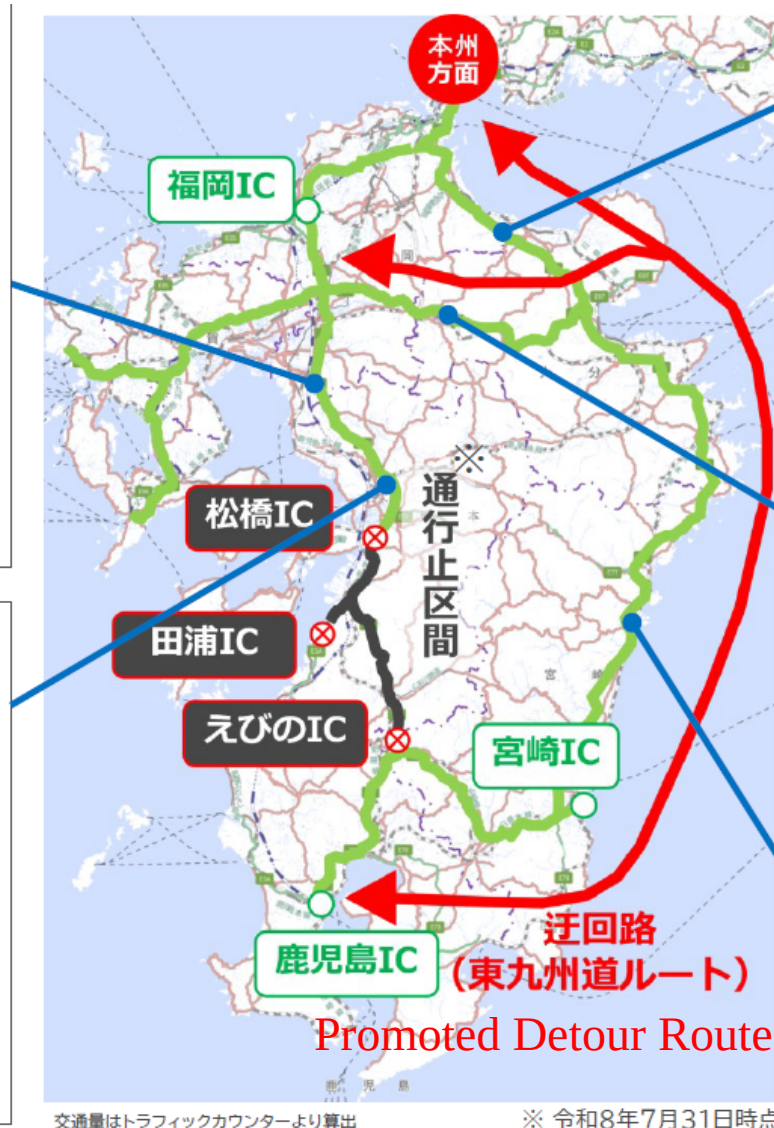
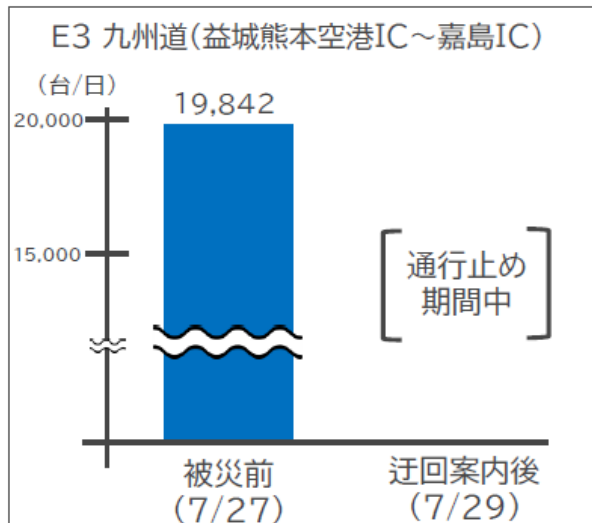
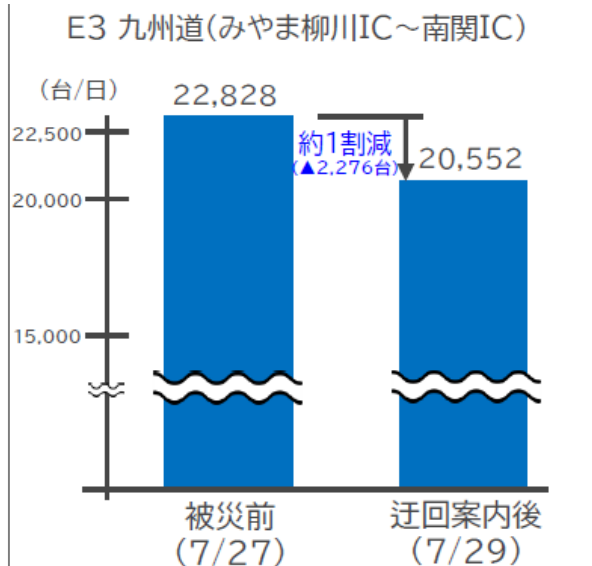


<https://www.nikkei.com/article/DGXZQOJC302SC0Q6A730C2000000/>

NEXCO WEST

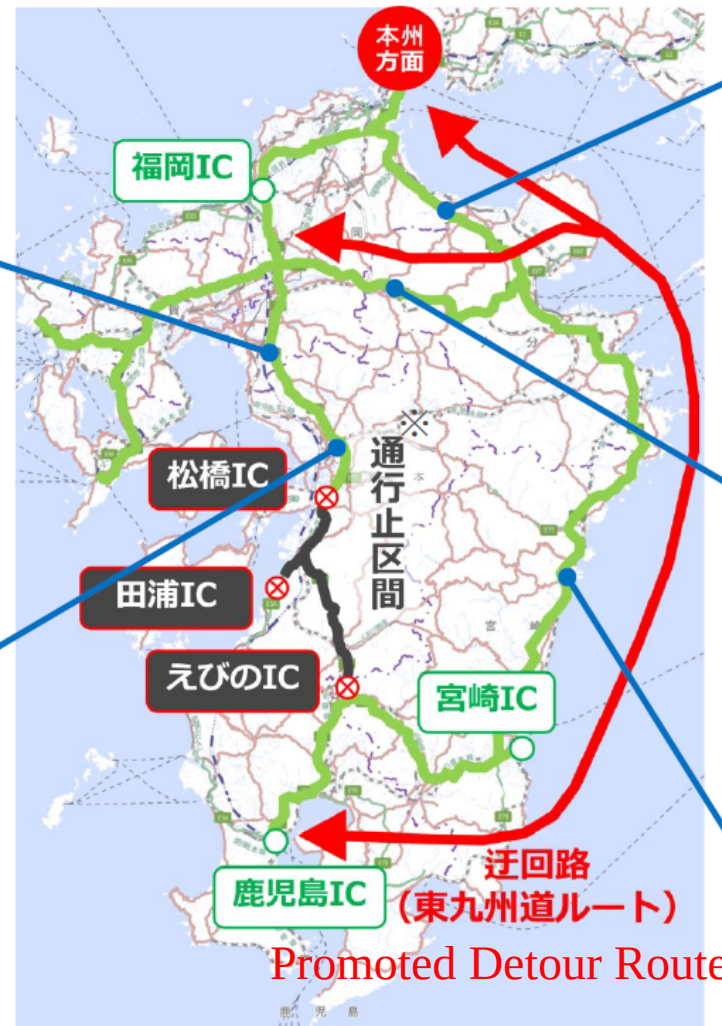
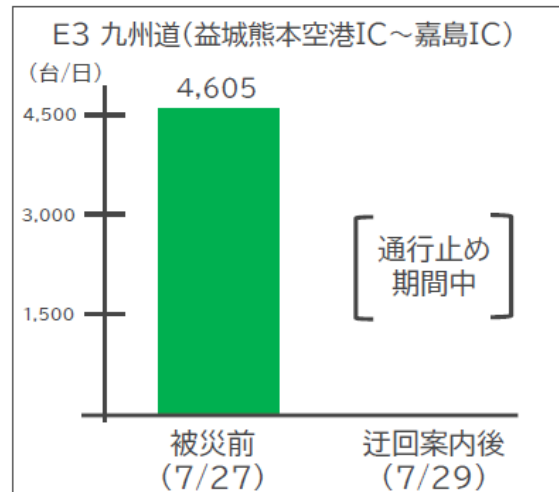
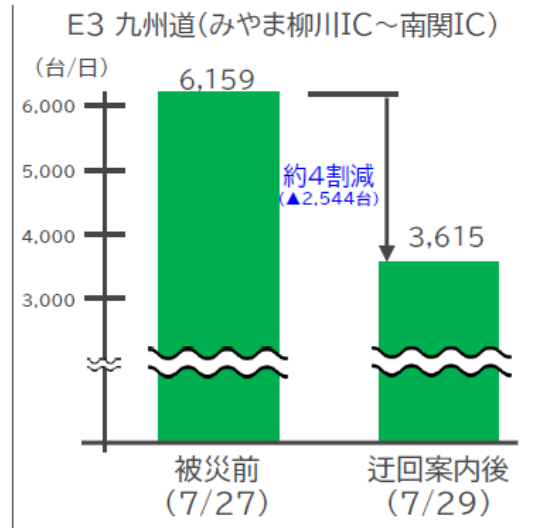


Promoting Wide-area Traffic Detours



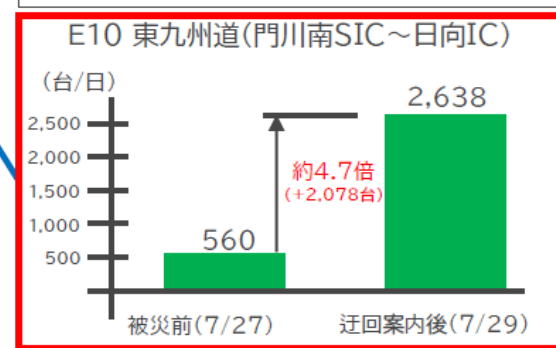
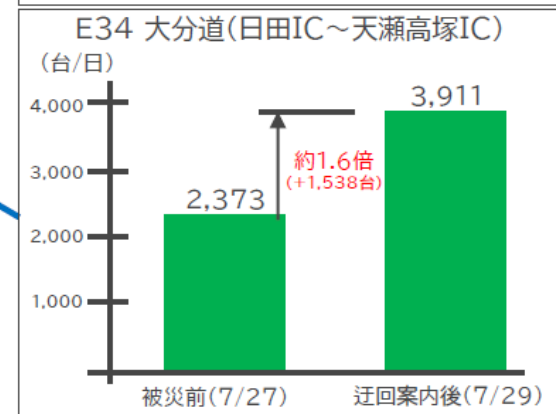
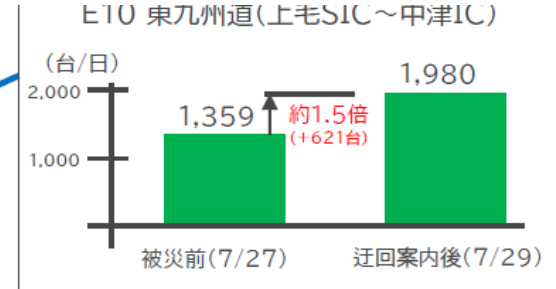
Changes in traffic count (July 27 → July 29)

Promoting Regional Traffic Detours (Trucks)

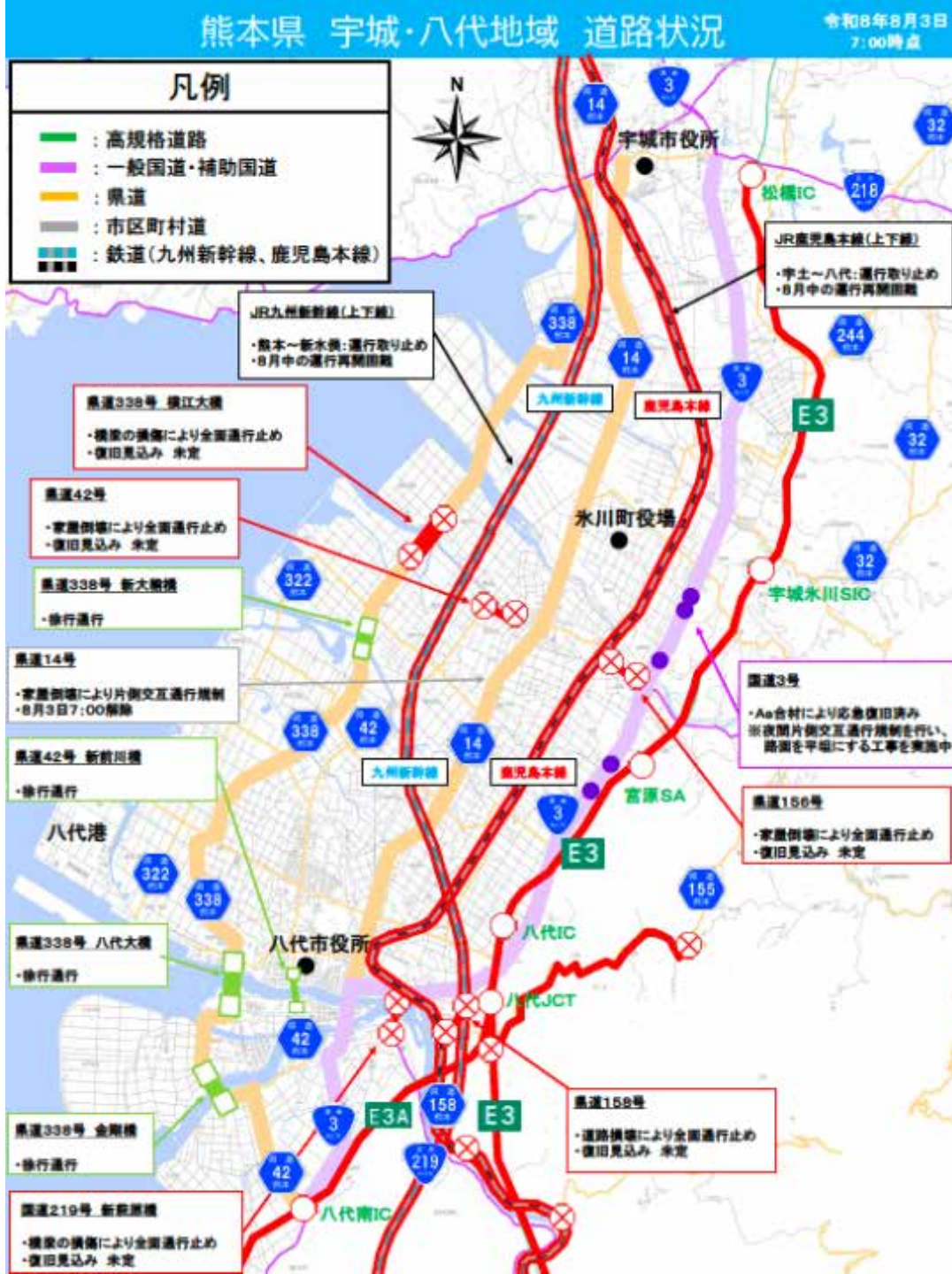


交通量はトラフィックカウンターより算出

※ 令和8年7月31日時点



Importance of Route Redundancy



Managing regional traffic in the affected corridor

August 3, 2026

MLIT

<https://www.qsr.mlit.go.jp/content/000002676.pdf>

<https://www.qsr.mlit.go.jp/content/000002689.pdf>



The corridor network

Normal Status:
2 Rail Lines &
4 Road Routes



Current Status:
Reduced to
2 Road Routes Only



August 2, 2026

One-lane alternating traffic restriction caused by collapsed building



Damaged bridge

Long time to repair



Managing Regional Traffic in the Affected Corridor



Speed reductions due to road damage

Operation made by the government (Aug 02)



**Step 1: Repair
& speed up**

One-lane restriction
→ Two-lane

Capacity recovery

Damaged
bridge

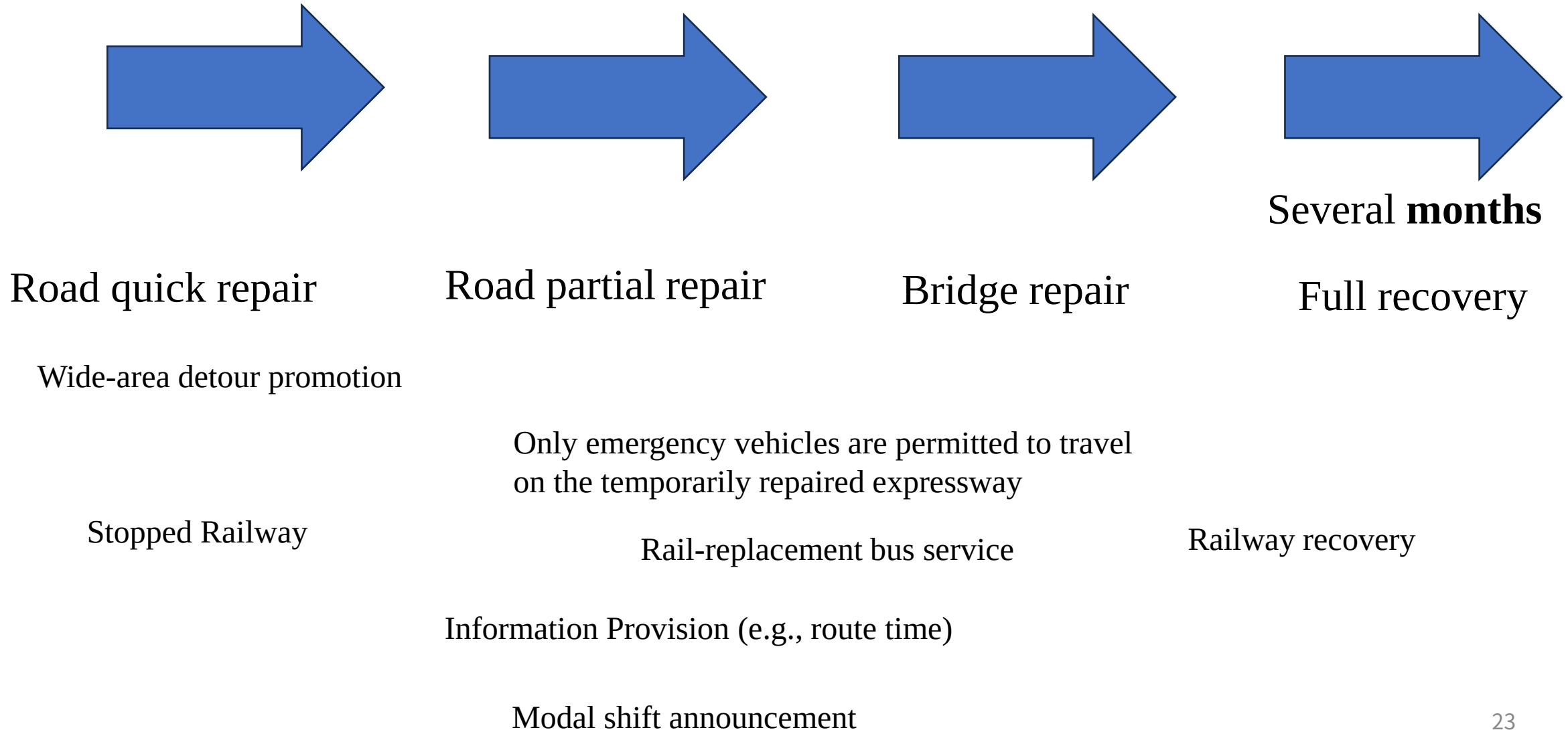


**Step 2: Promoting Detour
Route change**



**Step 3: Repair
& speed up**

Anticipating **Phase** Changes (Examples)



Collaboration among many players in disaster traffic management

- Government road-division (local, national)
 - Government public transportation division (local, national)
 - Police
 - Railway company, bus company, taxi company
 - Trucking association, Bus association
 - University Researchers
-
- Communication in normal situation will help

**A 10-year recovery process
from the 2016 Kumamoto Earthquake**

**Key lessons learned
for the 2026 Earthquake**

Damage Overview by the 2016 Earthquake in Mashiki

Human damage (unit : person)

Victims	20
Related death	20
Seriously injured	132

(2017.6.20)

Housing Damage

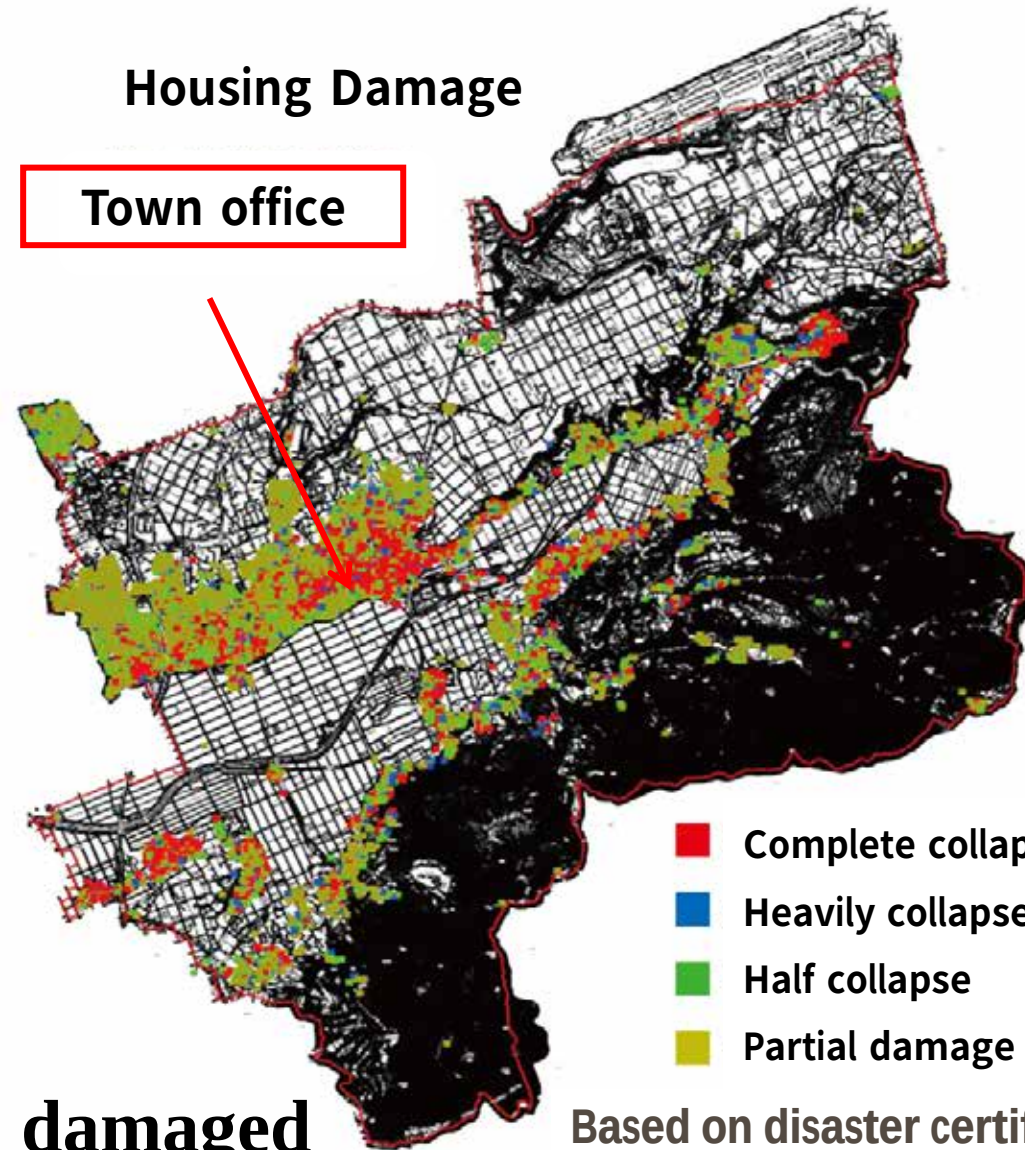
(unit : house)

Complete collapse	3,026
(Heavily) Half collapse	3,233
Partial damage	4,325
Total	10,584

(2017. 6.20)

Housing Damage

Town office



- Complete collapse
- (Heavily) Half collapse
- Half collapse
- Partial damage

Almost all the housing units were damaged

Based on disaster certificate

Damage by the 2016 Earthquake in Mashiki



Shimo-machi



Terasako



Miyazono (photographed on rooftop of town office)

Disaster Recovery Planning



Frequent Public Meetings to Capture Local Perspectives

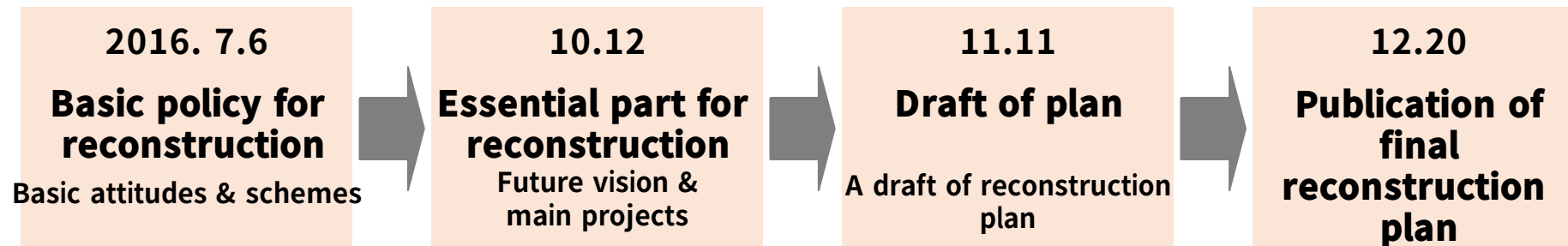
Local Town Meetings (21 sessions), Meetings at Temporary Housing Sites, Resident-Wide Questionnaire Surveys, Surveys for Elementary and Junior High School Students, Workshops for Youth , Public Comments, etc.

Disaster Recovery Planning

Reconstruction Planning

Committee for reconstruction planning & technical sub-committee discuss the reconstruction plan

member...local people, scholars, prefectural officer, prefectural assembly, town councilors, several groups in town
Advisory board: Emeritus professors, MLIT and several organization etc



Town Office



Original office in 2016



Renewed office

No damage in the 2023 earthquake
(Few residential housing damage)

Mashiki has become a disaster-resilient town

Road widening project





Road Widening Project

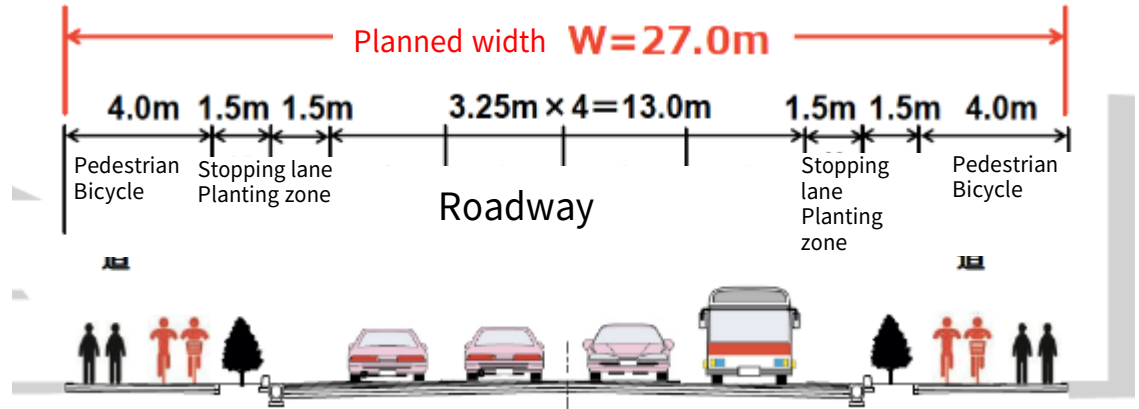
sectional view

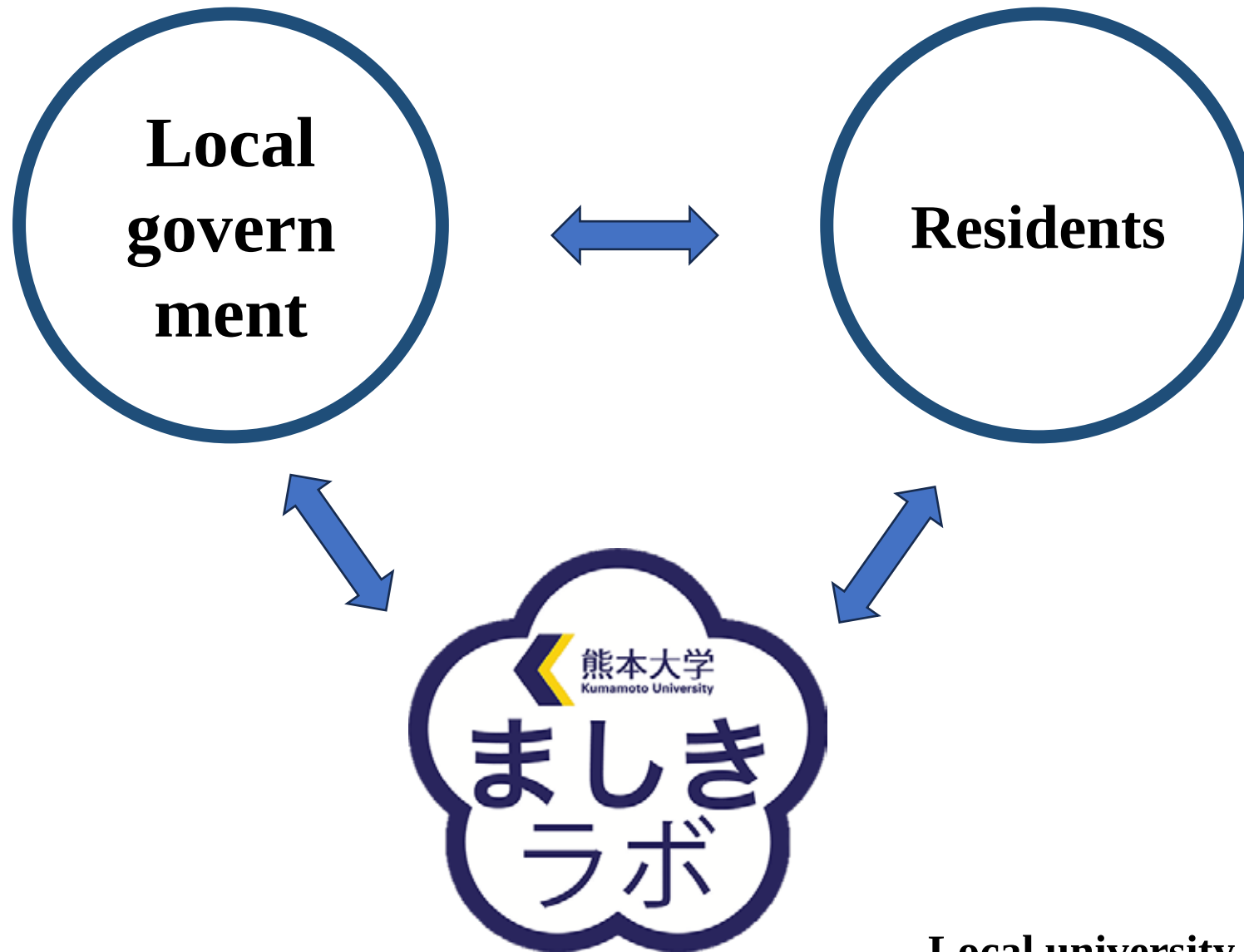
Previous width

$W = 10.0\text{m}$



Planned width $W = 27.0\text{m}$





Local university as a mediator

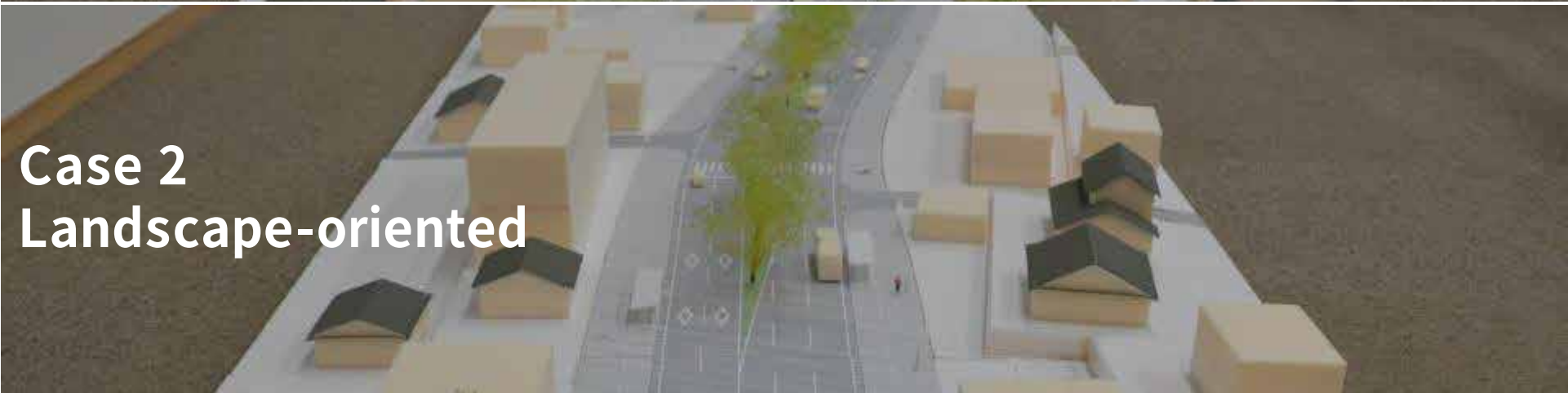


Opinions from mothers

**Case 1
Normal**



**Case 2
Landscape-oriented**

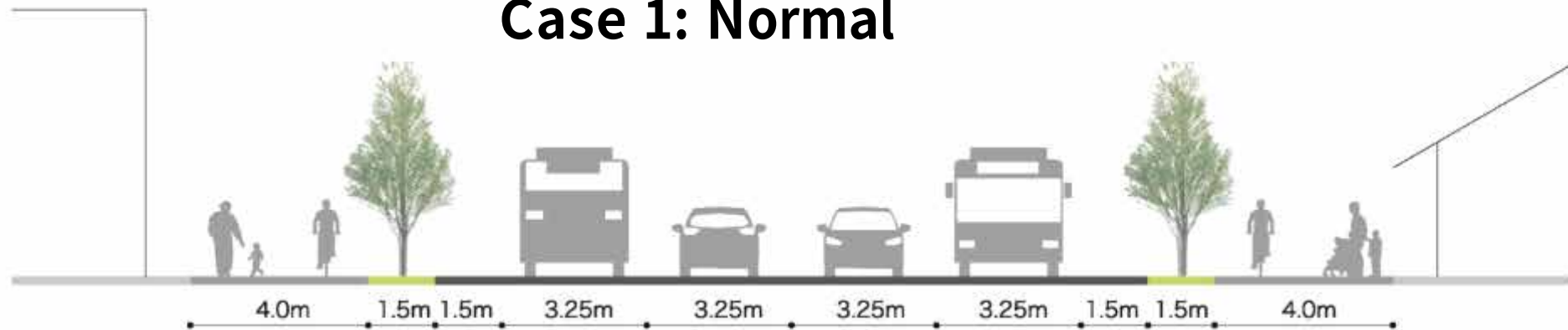


**Case 3
Roadside env.**



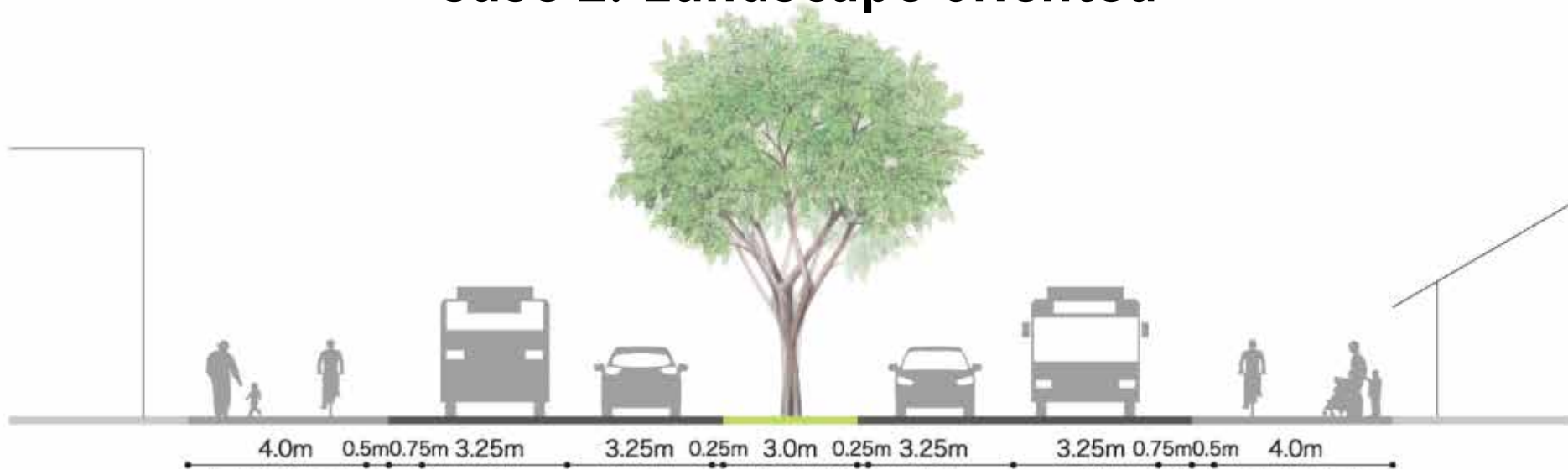


Case 1: Normal



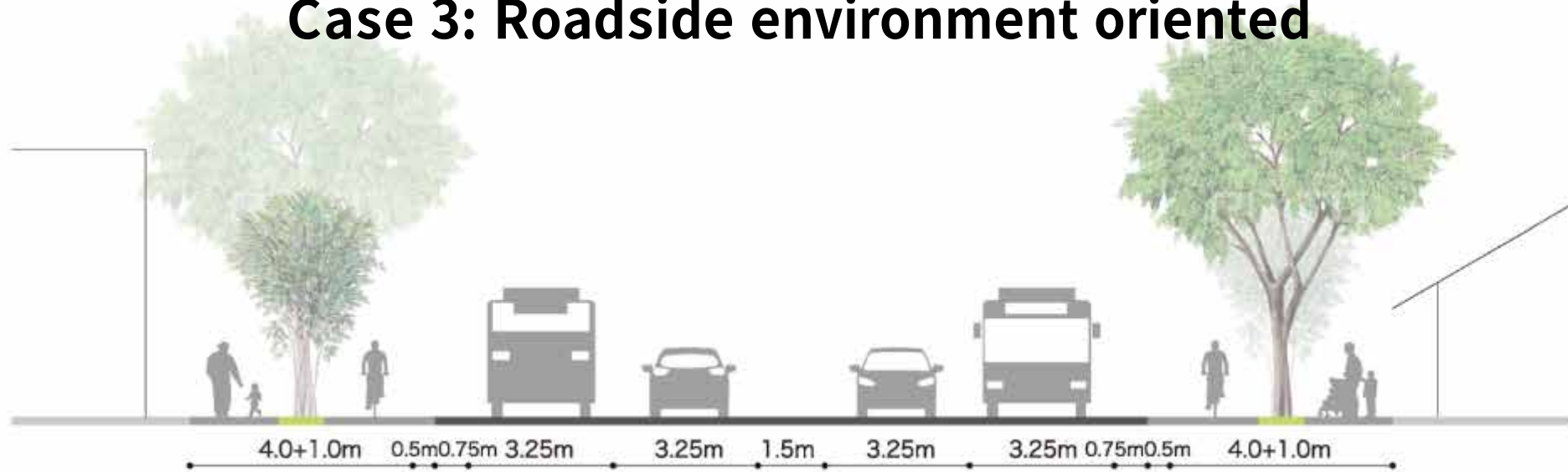


Case 2: Landscape oriented





Case 3: Roadside environment oriented





Case 1



Case 2



Case 3

Walker



Bus stop



Roadside



Crossing



Landmark





Meeting with youth





Submitting Recommendations on the Road to Governor

August 24, 2018



Most recommendations were realized in the end



Field experiment (Oct 2, 2019)





Designing the details of bus stop (June 2023)



Road Opening in March 2026: Commemorative Walking Event



Road Opening in March 2026: Commemorative Walking Event



Road Opening in March 2026: Commemorative Walking Event

Lessons learned

- **Local university can serve as a mediator between residents and local government**
- **Citizen participation is important**
- **Continued support can be made possible by local university**
- **Anticipating phase change**
- **...and many**

Concluding remarks

- **Disaster traffic management (Short-term ~ several month)**
 - Collaboration with many players
 - Measures can be simple: Information provision, route change promotion, modal shift, peak shift, etc
 - Fields visit is important: Understanding pavement situation
- **Disaster-resilient transportation network (Long-term ~ several years)**
 - Redundant transportation network is important
 - Citizen participation & mediating players are important for planning
 - Developing land use & transportation network considering disaster risk (e.g., active fault & disaster hazard)

Appendix

The Int. joint symposium & WS on TOD 2026
Univ. of Seoul, Aug 4, 2026

A postdoctoral researcher at UT Austin 20 years ago



July 2007



Nov 2007



March 2007

Takuya Maruyama
Kumamoto University, Japan

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Cases of the 2016 & 2026 Kumamoto earthquakes

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